Graphic Era

End Term (Odd) Semester Examination December 2024

Roll no. .....

Name of the Course and semester: M.Tech-CSE **Semester** 1st
Name of the Paper: Artificial Intelligence
Paper Code: MCS 142

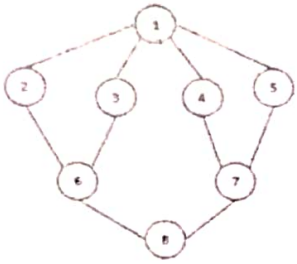
Time: 3 hour **Maximum Marks:** 100

Note:

(i) All the questions are compulsory.
(ii) Answer any two sub questions from a, b and c in each main question.
(iii) Total marks for each question is 20 (twenty).
(iv) Each sub-question carries 10 marks.

Q1. (2X10=20 Marks) (CO1)

a. Explain the difference between Artificial Intelligence and Natural Intelligence with relevant examples.
b. Analyze the importance of problem representation and state-space formulation in AI problem-solving. Use an example to illustrate your analysis
c. Solve a given problem using Breadth-First Search (BFS) and explain each step involved in the process.



Q2. (2X10=20 Marks) (CO2)

a. Explain the concept of predicate logic and how it is used to represent facts in AI.
b. Construct a semantic network to represent the following information:
   a. "A bird is an animal."
   b. "Penguins are birds."
   c. "Penguins cannot fly."
c. Evaluate the effectiveness of natural deduction as a method for reasoning in predicate logic. Use an example to illustrate its strengths and weaknesses.

Q3. (2X10=20 Marks) (CO3)

a. What are forward and backward reasoning? Briefly explain their significance in problem-solving.
b. Compare monotonic and non-monotonic reasoning, highlighting their differences and use cases.
c. Evaluate the effectiveness of the Dempster-Shafer theory in handling uncertainty compared to Bayesian Networks. Provide examples to justify your evaluation.

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Q4. (2X10=20 Marks) (CO4)

a. Explain how natural language processing (NLP) enables computers to understand human language.

b. Apply the perceptron learning algorithm to classify a set of linearly separable data points. Demonstrate with an example.

c. Analyze the limitations of traditional expert systems when dealing with ambiguous or incomplete data.

Q5. (2X10=20 Marks) (CO5)

a. Describe explanation-based learning (EBL) and discuss how it differs from traditional learning methods.

b. Compare and contrast learning by induction and explanation-based learning in terms of methodology and outcomes.

c. Evaluate the effectiveness of explanation-based learning in domains requiring deep reasoning, such as medical diagnosis. Provide examples to support your argument.